Investigation of seismo ionospheric anomalies associated with the Mw 6.7 Mongolia Earthquake

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Dr. Munawar Shah

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ABSTRACT

The seismo-ionospheric anomalies associated with the Mw 6.7 Mongolia Earthquake (EQ) occurred on Jan 11 ,2021, are investigated from the Global Navigation Satellite System (GNSS) based Total Electron Content (TEC) acquired from the stations in seismic region. To observe the EQ anomalies, it is mandatory to take into account the variations in solar and geomagnetic storm activities, the present study has the worldwide geomagnetic storm level, particularly the Disturbance Storm Time (Dst), Kp, Ap, F 10.7 indices prior to EQ and after the EQ. It is clear that the EQ happened during the quiet geomagnetic conditions Kp < 3. But the storm was active five days before the EQ. The electromagnetic variations within the ionosphere related to EQ are statistically examined in TEC from IGS stations. Our results from TEC clearly show that the ionospheric anomalies are induced several days before the EQ near the epicenter and positive anomalies appear on -4th Day, -3rd Day, -2nd Day, - 1 Day and on main shock day. Based on the obtained results, anomalies in TEC are more likely related to the geomagnetic storm indices due to its abnormality during the EQ preparation period. Furthermore, enhancement and depletion in TEC prior to EQ supports the presence of storm anomalies over the seismic zone.

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LIST OF ABBREVIATIONS

GNSS	Global Navigation Satellite Systems
PVT	Position, Velocity, Time
GPS	Global Positioning System
GLONASS	Global’naya Navigatsionnaya Sputnikova Sistema
BDS	BeiDou Navigation Satellite System
US	United States
MEO	Medium-Earth Orbit
GEO	Geosynchronous Earth Orbit
LEO	Low Earth Orbit
TEC	Total Electron Content
VTEC	Vertical TEC
EQ	Earthquake
IGS	International GNSS Services
USGS	United States Geological Survey
ISGI	International Services for Geomagnetic Indices
STEC	Slant TEC
UB	Upper Bound
LB	Lower Bound
UT	Universal Time

1 INTRODUCTION

All the problems associated with accurate positioning, timing and tracking are being resolved by satellite navigation.

1.1.Global Navigation Satellite System (GNSS)

Positioning Navigation and Timing (PNT) satellites have rapidly become a vital part of everyday life, although it is invisible but integral utility that is utilized by whole world and are also called as Global Navigation Satellite Systems (GNSS).

The SATNAV systems and their augmentations are defined by term GNSS. Global constellations within the GNSS, occasionally referred as fundamental constellations, nominally contain three or six orbital planes which are the path for four or more satellites per plane and in total twenty four satellites or more are arranged in these planes at an altitude of about 20,000km to 23,000km. Many countries introduced their own position navigation and timing services for their development in the beginning of 21st century. To achieve the accurate positioning is the main goal of GNSS. Human beings are searching for more innovative and radicle techniques for position, navigation and timing services since the early day. For hundreds of decades to figure out the location of different targets on the Earth's surface by utilization of different gadgets is experimented out. For instance, Positioning by means of stars and planets have been utilized for years and it is also termed as celestial navigation. Consequently, on a worldwide level the current developments in space technology have provided abundant chances to improve high-accuracy positioning and navigation systems. For reliable and accurate navigation different countries such as China, Russia and European Union launch their own navigation systems BeiDou, GLONASS and Galileo respectively are the pioneers who changed the theory of positioning to precision and reliable navigation on a global scale. All systems are collectively known as GNSS. As time is passing GNSS is continuously developing and achieving further consideration from the scholars and inventors [1]. From the 20th century onwards with the advent of satellite technology, the power of navigation has completely changed and now people will rely heavily on satellite technology for navigation purposes. Numerous countries are interested to participate in Earth's atmosphere and space with the development and growth in the field of science. Sputnik was the world's 1st satellite that is being launched by Soviet Union In October 1957.[2] Researchers started to work in the satellite positioning field by the Launch of Sputnik and later after the productive launch of Sputnik, Transit was launched. Earliest Transit navigation satellite was launched on October 1959. A new phase of position, navigation and timing technology had opened with the launch of Transit and it retired after 32 years of serving. Answers for the problems of navigation positioning and timing was provided by The Transit satellite mission, it worked on two frequencies 149.99 and 399.97MHz and it is on lower Earth orbit. Next was the turn for Russia. Cicada or Tsikada was the 1st navigation satellite that is being launched by Russia in 1974.it operates on same two frequencies 149.99 and 399.97MHz and have same orbit as transit had means it also operates in LEO. Later, after productive achievement of the Transit satellite mission, a new beginning for GNSS awaits on its horizon. In the next section will discuss different satellite navigation technologies one by one.[3]

GNSS include sections which are space segment comprises of orbital planes that are nearly circular that are placed at some altitude above Earth's surface and satellites revolves in these orbits. Following are the different orbital planes depending on altitude

1. Low earth orbit (LEO)
2. Medium earth orbit (MEO)
3. Geostationary earth orbit (GEO)

The altitude of low earth orbit is 2000km and satellites that are present in this orbit completes their one revolution around Earth in 95 to 120min. The altitude of medium Earth orbit is 5000 to 12, 000km and satellites that are present in this orbit completes their one revolution around Earth in six hours. The altitude of geostationary earth orbit is fixed at 35,786km and satellites that are present in this orbit completes their one revolution around Earth in 24hours which is the rotational speed of Earth because it is stationary with respect to Earth means it is at same position from Earth's view. Each satellite is equipped with instruments that are used for navigation or other extraordinary jobs. Information from ground control segment is being stored received and processed by satellite. Every satellite have different identification number or pseudo random number PRN to make them distinct from other i.e., system name, position in orbit etc. [4]

For the preservation, placement and maintaining the whole system there comes the name of Control segment which is responsible for all work. Moreover, this segment is also responsible for tracking of the satellites in their tracks and the clock parameters, observation of auxiliary data, and upload of the data message to the satellites. Further, data encryption and service security is also being provided by the control segment. Additionally, tracking stations that are present globally manage, control and monitor the activities using bidirectional communication between the control segment (GNSS stations) and space segment (satellite).

To get the information sent from satellite user segment is involved there which consist of different equipment to read the received messages from satellites mainly consist of passive receivers which are capable to decode information. Anybody can use receivers it is free of cost. However, GNSS military signals are prohibited to use by civilians. There are numerous GNSS receivers available in the market nowadays as shown in Fig. 1.1 [5]

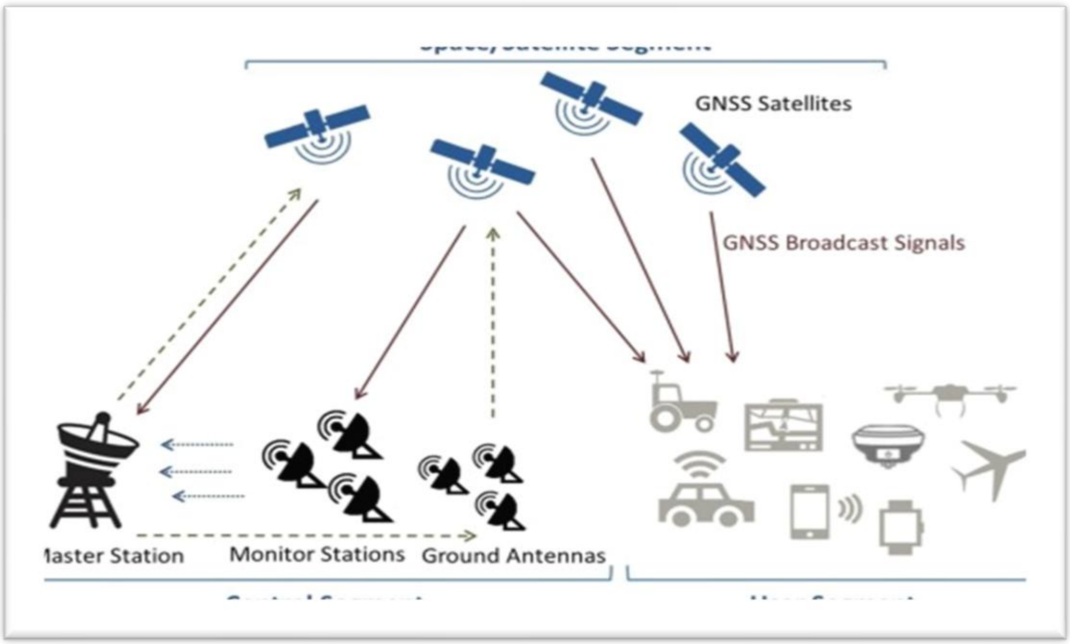

Fig. 1. 1 Shows three segments of GNSS [Courtesy: Sciencedirect.com]

1.1.1 History and Background of GNSS

"The Brick Moon" was the short story in which there is 1st time use of artificial satellite is proposed in 1869 that is being written by Edward Everett Hale[6].In astronomy perspective "satellite" term was 1st used by Galileo, to elaborate the moons of Jupiter he witnessed by his elementary telescope. In 1957 Soviet Union launched 1st ever artificial satellite that generates and transmits a signal that is received globally. Right after launch of Sputnik, U.S NAVY was financed for the mission "TRANSIT" by the US Department of Defense (DOD) in December 1958, its progress and launch began in 1959 and TRANSIT system become operational in 1964 as shown in (Fig 1.2) [7] . TRANSIT used six satellite that are at altitude of about 1000km in offset polar orbit and it transmitted the signal using Doppler shift. With the accomplishment of more innovative, up to date NAVSTAR GPS system in December 1996, the TRANSIT system retired. TRANSIT system was used by NAVY department. To determine the frequency modification, receiver in ship had to catch the satellite signal for ten to fifteen minutes. (Fig. 1.3)

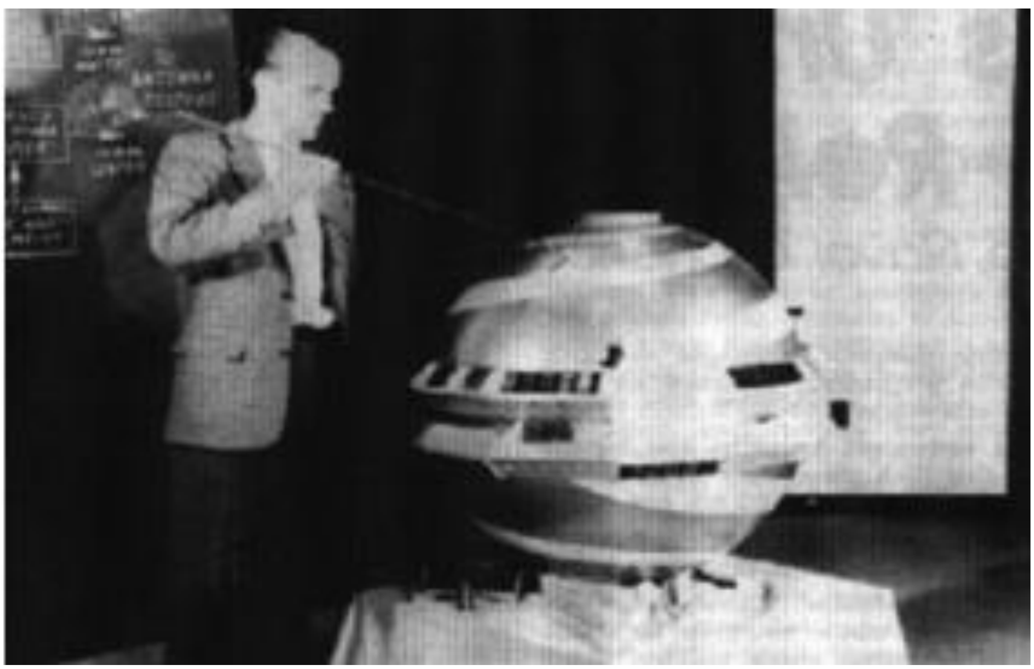

Fig. 1. 2 1-B TRANSIT satellite (Image courtesy of NASA)

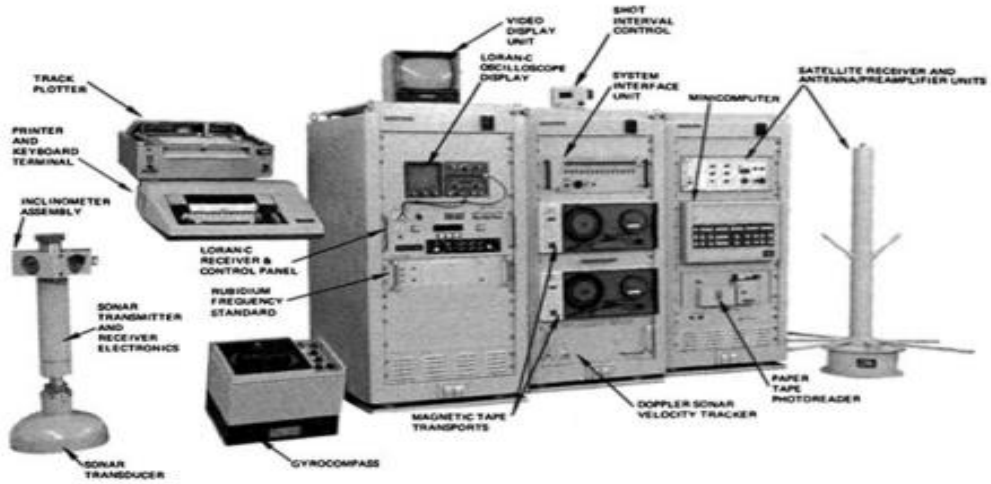

Fig. 1. 3 This figure displays a navy ship onboard non portable receiver system (The TRANSIT system).

Likewise, similar system “Tsikada/Parus” was being established by the Soviet Union rapidly that work on same Doppler shift approach. Soviet Union develops two systems that own similar designs and satellite but system used by civilians is Tsikada system and for defence purposes Parus was used. In 1978 “Tsikada” which was a civilian system is being launched and used by countless ships whereas, military constellation was being available in 1978 until is being replaced by GLONASS system. Then to overcome the limitations the new generation system has been developed that uses upgraded technologies and which is the beginning of NAVSTAR or GPS (Global Positioning System) [8] which is further discussed.

1.1.2 Components of GNSS

GPS (Global Positioning System) from U.S, GLONASS (Global’naya NAvigatsionnaya Sputnikova Sistema) from Russia and Galileo from European Union are the three core satellite technologies. They consists generally of three alike sections which collectively called Global navigation satellite system: (a) space segment, (b) control segment and (c) user segment.

1.1.1.1 Global Positioning System (GPS)

Global positioning systems (GPS) was initiated in 1973, and in 1993 it attained IOC. It entertain its users globally without any fee with precise positioning, navigation and timing information[9]. As of June 15, 2021, there were a total six orbital planes comprised of 31 operational satellites in the GPS constellation. The satellites orbits at an inclination of 55 degrees to the equator and complete their revolution around the earth in 12 hours, i.e., revolves twice around the Earth in a day. Two ranging codes is sent by GPS satellites C/A code and Precision code where C/A code is used by civilians and P code is used for defense purposes. It has five frequency bands $L1 - L5$ where L1 and L2 are used and L5 is reserved for Safety-of-Life. The signal consists of GPS data and time, ephemeris data and the almanac. L3 band is used for nuclear weapon detonation, detection and treaty or ban enforcement. [10]. Tracking GPS satellites, monitoring their transmission, checking integrity, analysis performance, send commands and data to the satellites is done by control segment depicted in Fig (1.4). Ground network comprises of one master control station, one alternate master station, twelve command and control antennas, and 16 monitoring sites. User segment comprises of receivers and antennas that catches and receives the signals and further calculate position, velocity of satellite, ephemeris and almanac data present in these signals. To get accurate positioning receiver has to acquire and maintain lock on four satellites by a technique called trilateration[11].

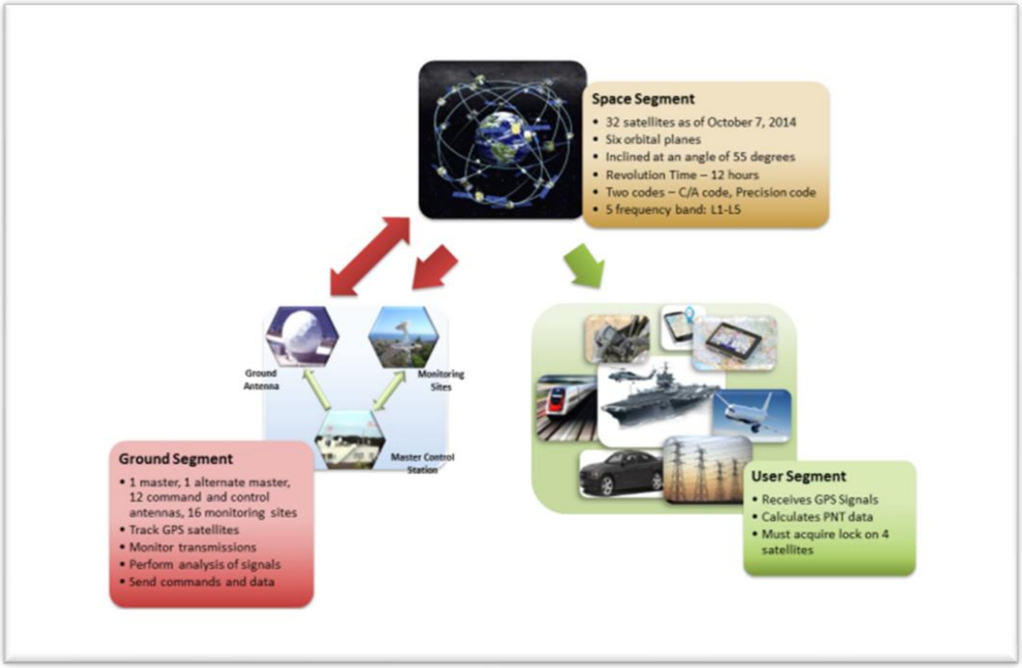

Fig. 1. 4 GPS space, ground and user segment

1.1.1.2 GLONASS

The Russian federation launch their 1st satellite in 1982. Their system is known as Glonass system ("GLObalnaya NAvigatsionnaya Sputnikovaya Sistema," or GLObal NAvigation Satellite System) was commenced on 1976. Space segment of GLONASS comprises of three orbital planes having eight satellites in each plane and in total twenty four satellites are there. Glonass unlike GPS uses FDMA technology which uses CDMA technology, transmits signals on two ranging codes C/A that is used by civilians and works on L1 band and P-code that is utilized by military and works on two frequency bands L1 and L2. Russian federation innovate their GLONASS mission after 2000 and launched their GLONASS-M satellite On December 10, 2003 although it doesn't acquire its preferred results and as a result this satellite mission was terminated. Later on, GLONASS modernized their program by shifting their frequency on CDMA from FDMA to entertain their users with better positioning and navigation and launch GLONASS-K [12][13].

Table 1. 1 GLONASS upgradation

1982 (GLONASS)	**2003 (GLONASS-M)**	**2011 (GLONASSK1)**	**2014 (GLONASS-K2)**
3 year life design	7 year life design	10 year life design	10 year life design
Clock stability $5*10^{-13}$	Clock stability $1*10^{-13}$	Clock stability $\sim5*10^{-13}$	Clock stability $\sim1*10^{-14}$
Signals: L1SF, L2SF L1OF (FDMA)	GLONASS+ L2OF (FDMA)	GLONASS-M+L3OC (CDMA)	GLONASS-M+L1OC+ L3OC+L1SC+ L2SC (CDMA)
Total 81 satellites launched	Total 36 satellites launched by 2011	SAR	SAR

1.1.1.3 GALILEO

GALILEO satellite mission was being launched by EU which provide their civilians with high accuracy and guaranteed positioning services. Galileo space segment comprises of three medium earth orbital planes inclined at 56 ° with height of about 23,222km with thirty satellites revolving in these planes [14]. Galileo works on three frequency bands E1, E5 and E6 and strength of E1 signal frequency is similar to GPS L1 signal. [15].

Summary of GNSS constellation is illustrated in following table [1].

Table 1. 2 GNSS constellations

System	**GPS**	**GALILEO**	**GLONASS**	**BEIDOU**
Number of satellites (in orbit)	31 (24)	28 (24)	24 (18)	27MEO 3IGSO 5 GEO
Spare satellites	8 active	3 active	3 planned	3; 0; 0 planned
Status	Fully operational	Fully operational	Partially operational	22 satellites operational
Orbit	6 MEO	3 MEO	3 MEO	3MEO 2 IGSO 1 GEO
Orbital altitude [km]	20180	19130	23222	23150 35816 35816
Orbital inclination [º]	55	64.8	56	55
Revolution period	11h 58mn	11h 16mn	14h 07mn	12h 38mn

1.2. GNSS Applications

Information derived from GNSS signals is used in many ways. Those uses are known as applications. These applications are known as tracking or surveillance. Air applications include if device moves above the Earth and it has installed GNSS receiver within. From helicopters to airplanes to spaceships and even to satellites, GPS (and soon other GNSS), GNSS is broadly assist lot in navigation, positioning, tracking, aviation operations, sensor annotation, science, recreational activities and communicational abilities [12]. Some of its applications includes Transportation , Aviation , Agriculture and fisheries , Civil engineering and Energy [16]. To track the position of vehicles and engines, maintenance vehicles etc., GNSS collaborate with other technologies for that purpose. GNSS also serves to figure out the exact positions of trains to get rid of accidents, transportation delays their cost and in the end customer services. In aviation, the departure, route and landing of airplane is being monitored by GNSS. Areas which are isolated and not facilitated with navigation helps there GNSS assists aircraft navigation and it is a noteworthy factor of collision avoidance systems, and of systems helped to advance methods to airport runways. In marine transportation, when ships are in open sea or when they try to maneuver in narrow ports GNSS assist in determining their positions. PNT is integrated into underwater surveying, buoy positioning, navigation hazard location, searching, and mapping. Agriculture and fisheries is another application of GNSS that includes accurate chemical spraying, Crop production observation, Navigation and observing fishing vessels[17]. Engineering includes Structures observation, Equipment management, transport ways and Railway track supervision etc. as shown in Fig (1.5)[18].

Fig. 1. 5 Application of GNSS (a)transportation (b) Agriculture (c) Marine (d) Aviation

1.3.Motivation and objective

Purpose of GNSS launching not only includes navigation and positioning but it also includes earth and space observation, and to study and monitor the upper atmosphere of Earth GNSS-TEC technique assist in performing that job. This includes ionospheric disturbance, solar flares, solar storms etc. Ionosphere is the medium that contains free electrons and signals that are EMT waves generated onboard are influenced and hindered when they travel through ionosphere. This time delay could be used to deduce the changes in ionosphere by Total Electron Content (TEC). This work in this thesis is the still ongoing debate about the possibility of earthquake precursors. Researchers promoted and published results of the several earthquakes showing the existence of TEC enhancement or depletion in ionosphere before and after the earthquake [19][20][21]. Also, the geomagnetic storms, atmospheric circulations and magnetic activities are sources of ionospheric perturbations .In this study we checked geomagnetic activities in EQ days to get to know whether ionospheric anomalies are due to geomagnetic activity, solar storm or EQ. The aim of this thesis is to collect statistical information present in the ionosphere. To gain this information, TEC calculated from different GNSS stations measuring signals from satellites were used. The objective is to quantify the changes in TEC from different stations and analyze them in order to approach required results. Aims in this thesis includes:

- Find a suitable method of extracting TEC data from RINEX files from the data taken from GNSS Stations.
- Investigate geomagnetic and storms prior to EQ.
- Differentiate either ionospheric anomalies are due to geomagnetic storm or EQ.

1.4.Layers of atmosphere

Sun is the source of energy on surface of Earth in the same time it also ejects harmful and injurious short wavelength radiations also our space contains a lot of comets and asteroids that damage the earth if they reach at its surface. Earth's atmosphere consist of different layers that secure the planet from all the dangers of the space. The lower layer of the atmosphere is considered to lie on Earth's surface, the upper layer exist in space. The atmosphere of Earth is distributed into four layers as shown in Fig. 1.5.

- Troposphere
- Stratosphere
- Mesosphere
- Ionosphere

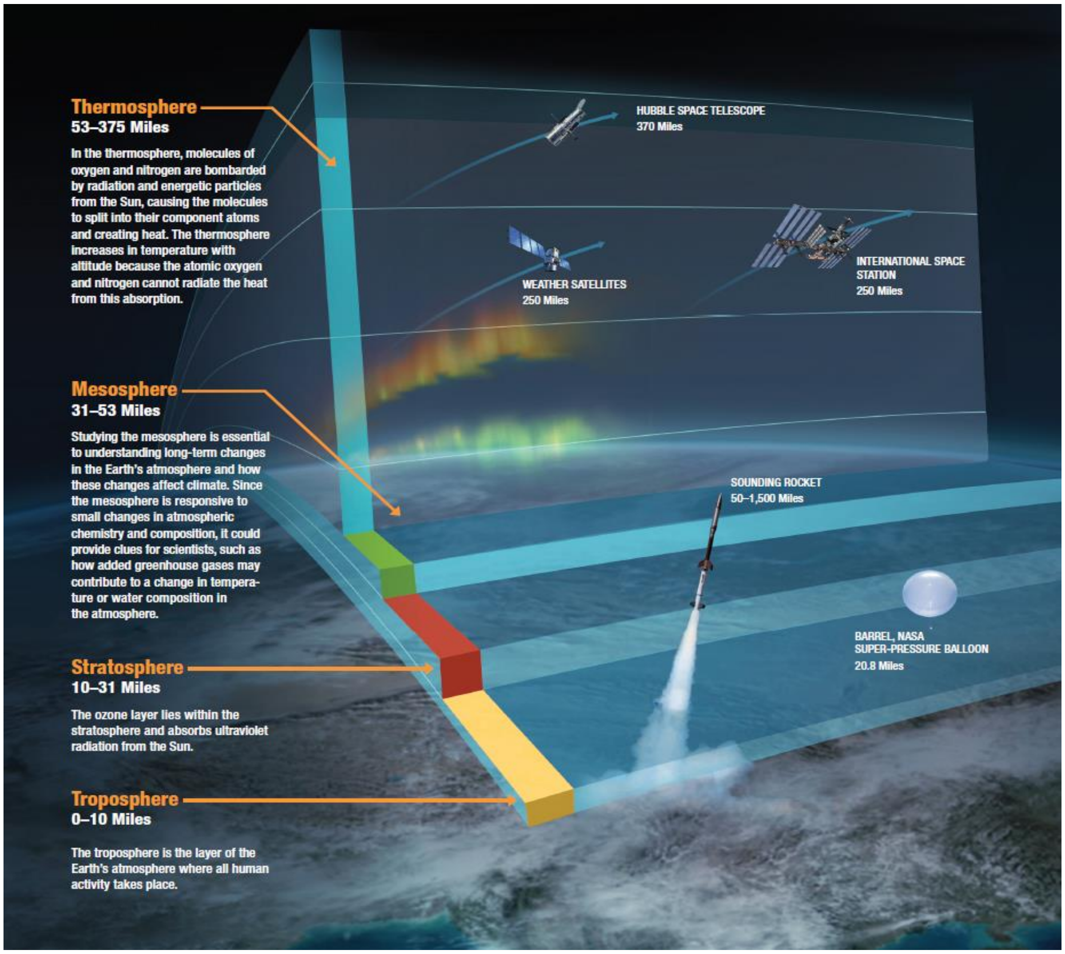

Fig. 1. 6 Layers of Earth's atmosphere (Image courtesy of nasa.gov).

1.1.3 Troposphere

Very 1st layer of Earth's atmosphere is troposphere that comprises of volcanic gases, pollution and also consist of weather system. As the distance increases from warm Earth's surface it has been observed that temperature in this layer decreased as you go upward. Therefore, the thickness of the troposphere is more at equator that is 16km and less at the poles that is 8km. There also exist boundary between troposphere and the next overlaying layer which is known as tropopause. -50˚C is typical temperatures at this altitude. The tropospheric region is a common source of atmospheric convection (warm air rises above cold air) it should be noted and which is crucial cause of gravity waves and gravity waves normally propagate perpendicularly and parallel away from their cause and as the background density reduces its amplitudes rise with height .The structure and dynamics has been spontaneously modifies as waves travel upward and therefore it causes disturbance in the other atmospheric regions.

1.1.4 Stratosphere

Right above the troposphere there exist another layer of atmosphere that is second layer of Earth's atmosphere titled as stratosphere. The stratosphere ranges from about 13 miles to its upper border the stratopause about 30 miles [50 km]. Ten kilometers is a high belt where one cannot live without oxygen. Ozone layer is also present in stratosphere that protects whole planet from harmful radiations emitted by sun. The upper two thirds of stratosphere contains higher concentration of ozone molecules that absorbs ultraviolet solar radiations and hence air density decreases that results in increase kinetic energy that means temperature increases upward in the stratosphere. High temperatures approaching 0˚ C in the stratopause separating the stratosphere and the upper mesosphere.

1.1.5 Mesosphere

Above stratopause there exist another layer of atmosphere that is mesosphere .It is a region of that extends from 50km to 80 km almost. It is coldest region of earth's upper atmosphere as altitude increases temperature decreases at this level to 263.15°K(-10°C) reach at maximum of -90°C at about 80km height. There exist a border among two regions of upper atmosphere called the mesopause.

1.1.6 Ionosphere

Another layer of earth's upper atmosphere after mesopause is ionosphere which is partially ionized that ranges from about 60 km to 1000 km, photo-ionization took place due to the solar emissions like EUV and X-ray radiations. The segment of this layer of the atmosphere consisting of several sub segments which are 1. Mesosphere, 2. Thermosphere, and 3. Exosphere also called as the ionosphere, and plasmasphere exist right above to ionosphere. The electromagnetic waves are highly influenced by free electrons that are densely present in this layer and is therefore called as ionosphere. Also, there exist a phenomena called ionospheric refraction which is defined as "The free electrons and ions influence the transmission of radio wave signals i.e. electromagnetic waves". It should be keep in note that there is constant variation in the structure of the ionosphere, i.e. its structure varies with day to night, continuously change throughout four seasons, latitude, during solar activities, and solar cycle phase.[22]Further, this atmospheric layer is also divided in four chief segments that are D, E, F1, and F2. Because of the recombination process these segments are separate which entirely rely on density of atmosphere which shows variation with height as depicted in Fig. 1.7

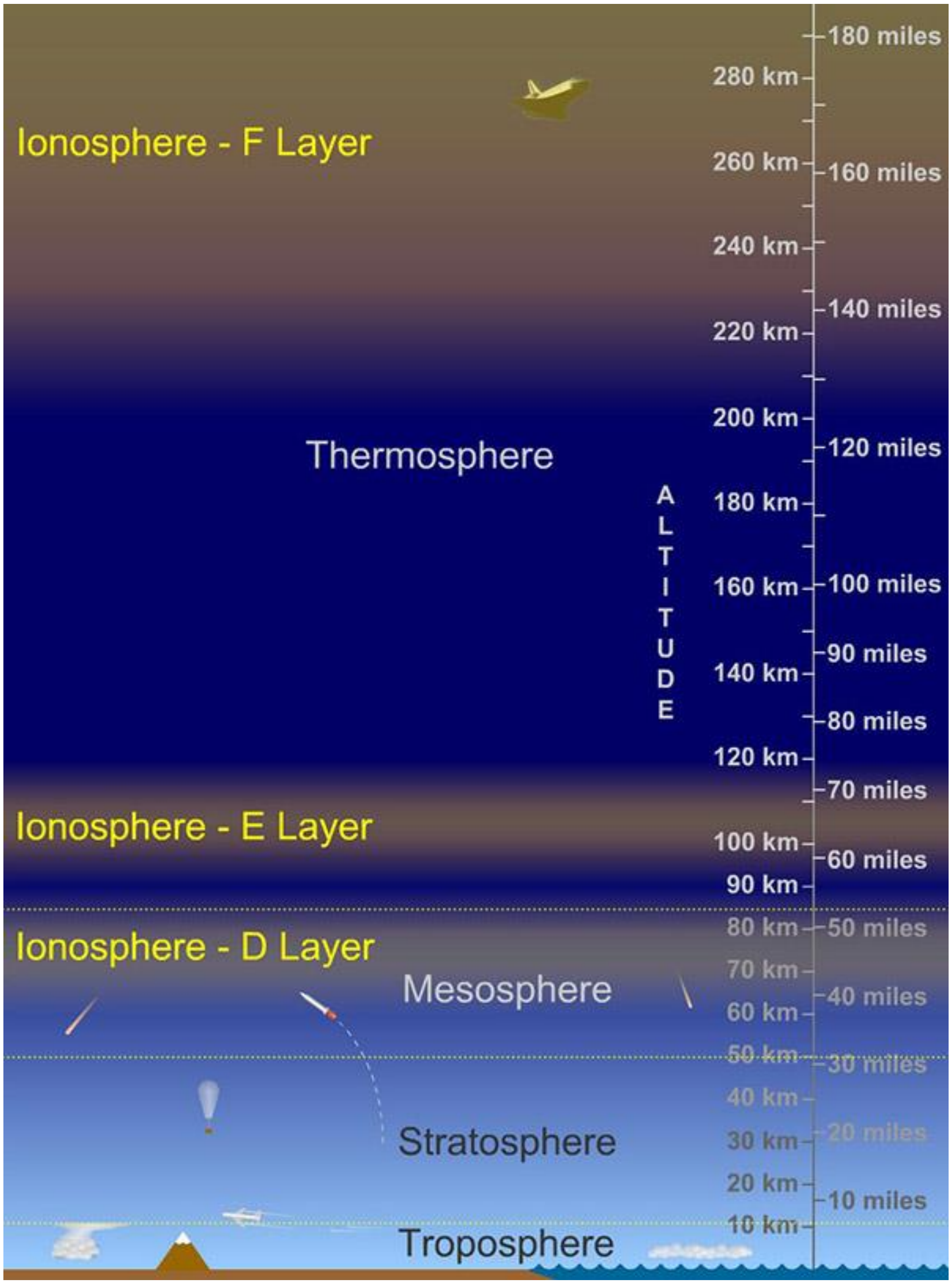

Fig. 1. 7 Different regions of ionosphere

1.1.6.1 The D region

Already elaborated above that ionosphere is further classified along altitude into different segments one of them is D layer and that D layer ranges from altitude of about 50 km to

90 km. It tremendously reduces after sunset. The electron density values are around 102 to 103 el/cm^3. There exist another layer beneath D layer that is C layer. [23]

1.1.6.2 The E region

After D layer there comes another region of ionosphere that is E region. E layer's characteristics depends almost entirely on two factors that are level of solar activity and sun's zenith angle. The interesting fact about the E layer is that unlike D and E it is present only at day time. [24]

1.1.6.3 The F region

The F layer that is also titled as Appleton layer is the topmost layer of ionosphere and located at about altitude of 140 to 1000 km. The major peak electron concentration lie at an altitude of about 250-300 km and the chief causes of ionization in ionosphere are Extreme Ultraviolet (EUV) solar radiation having wavelength 20 to 90nm. During the existence of sunlight at the day time F layer is distributed into sections, named as F1 and F2 moreover, F1 layer is lower layer during day time and have the range of about 140 to 300km above Earth and F2 is upper layer. F layer consist of only one layer at night. Interesting fact about F1 layer is that occurs merely through daytime and vanishes at night. Extreme concentration of F1 layer exist just after noon, local time, when the sun is straight above.[24]

1.5.Ionosphere and GNSS

Ionosphere is an inhomogeneous medium which is proved when some challenges regarding fading signals reflected by ionosphere were faced by radio/communication signals. Additional studies involve that ionospheric irregularities can cause focusing and defocusing of radio waves or you say it can cause fading of GNSS signal. Additionally, ionosphere medium can alter the traveled pathway of radio signal towards the Earth when receiver that generate radio signal transmits signal below ~30MHz frequency , this specific effect provides possibility of long-distance communication. On the other hand, there exist an instability in power of signals, propagation delay, weakening of signal and its degradation and in severe circumstances loss of catching the signal when receiver that generate radio wave transmit signals at higher frequency, and these effects has established momentous dangers to both communication and navigation systems because of existence of ionospheric plasma. Total electron content is being stated as the quantification of the EMT signal influence is obtained by integrating the electron density along the way that a signal track from space constellation to user end. TEC is defined as an integral of electron density along the path between the GPS satellite and the receiver, where 1 TECU = 10^{16} electron/m^2.

1.6.Effects of Ionosphere on propagation of radio signal

1.1.7 Refractive index

The refractive index n_r of the medium is dependent on velocity of electromagnetic wave in a medium and speed of light in vacuum, which is well-defined by following equation:

$$nr = \frac{c}{v} \qquad (1)$$

Where v depicts the propagation velocity in the medium and c depicts the speed of light in vacuum. Frequency, phase velocity and group velocity doesn't influence the speed of propagation of pure (unmodulated) wave in a non-dispersive medium whereas, frequency phase velocity and group velocity influence the speed of propagation of pure (unmodulated) wave in a dispersive medium. Phase travels faster than the speed of light in vacuum according to Eq 1.

1.1.8 Ionospheric delay

Ionosphere changes its behavior during the day and during the night time. During the day timings solar radiations ionize the neutral atoms present in ionosphere which results in production of ions and free electrons. Solar radiation is the core cause for increasing density of electron in ionosphere and during the night time recombination processes take place when free electrons recombine with ions present in ionosphere and they again produce the neutral atoms and hence the electron density decrease during night time and this way Electron density in ionosphere influence the propagation speed of the radio waves or GNSS electromagnetic signals. Ionospheric delay is produced which is the key error source and produce the error that exceeds 100m in extreme ionospheric conditions in positioning, navigation and timing (PNT). There are different models designed to overcome the effects of ionosphere delay and its effects on GNSS signals like Klobuchar model, NeQuick model and Neustrelitz TEC Model (NTCM) etc. can be utilized for real-time or post-processed applications.

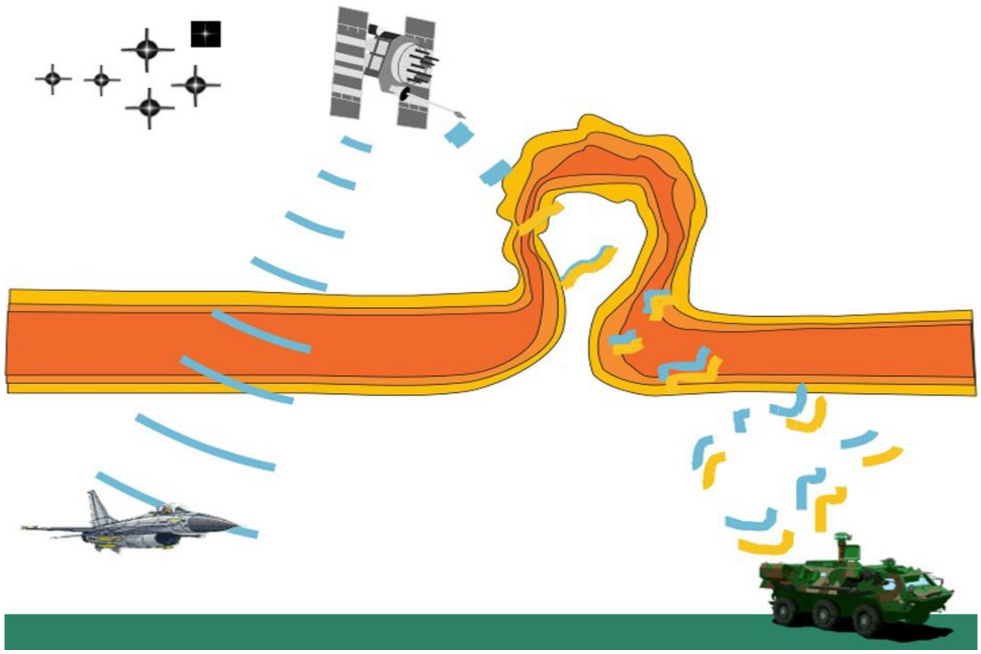

Fig. 1. 8 I*onospheric* irregularities affecting GNSS signals

1.7.Ionospheric variations and anomalies

The ionosphere is a very dynamic medium. As mentioned before Sun radiation is the main source of ionization and therefore introduces periodical variations in the ionosphere related to the time of the day or season. Solar disturbances or geomagnetic field disturbances can cause ionospheric perturbations. There are also other non-periodical but reoccurring variations or anomalies of the ionosphere. Ionospheric storms signify an extreme form of 'space weather', which can have momentous, adverse effects (restriction or reduction or totally blackout of power supply due to the production of high energy particles in satellites that created the destructive currents in grid stations, greater threat of radiation exposure on constellations and high altitude aircrafts, errors generated in GNSS systems and VLF navigation systems, harm to HF communications, disorder of UHF satellite links on ground and space based technological systems that are increasingly crucial to governments, corporations, and the lives of ordinary citizens of our globe.

1.1.9 Geomagnetic storm / ionospheric storm

Sun is ball of fire which emits energy in the form of radiations which are absorbed by different atmospheric layers of Earth. Magnetic storms in ionosphere are caused by solar activities which cause disturbances and variations in ionosphere which is very exposed from outer space and these storms are generally called ionospheric storms. Solar/interplanetary plasma arrive at high speed to magnetopause that is beginning of geomagnetic storm and leads to Solar wind energy in magnetopause. The storm-time neutral winds can alter the F region electron density peak height by pushing ions and electrons upwards and downwards along magnetic field lines. During magnetic storms, which mainly happen while the solar activity is increased, the emissions produced by Sun interacted with the Earth's magnetic field to create this phenomenon of rare beauty called aurora. To study ionospheric storms geomagnetic indices are introduced. Geomagnetic storm indices are Kp, Ap, DST, AE.

1.1.9.1 Solar-Terrestrial Indices

Geomagnetic disturbance level is specified by geomagnetic indices. Ground based equipment like geomagnetic stations are located all over the globe to monitor geomagnetic indices. Geomagnetic indices are Ap, Kp, Dst and AE. Geomagnetic classification according to DST, Kp, Ap and AE is shown in Table 1.3, 1.4 and 1.5.

Table 1. 3 Geomagnetic classification according to DST index

DST value	Storm type
Minimum Dst below -20 nT	Weak
Minimum Dst below -50 nT	Moderate
Minimum Dst below -100 nT	Strong
Minimum Dst below -200 nT	Severe
Minimum Dst below -320 nT	Great

Table 1. 4 Kp- index

Kp-index	Storm type
0	Quiet
1	Quiet
2	Unsettled
3	Unsettled
4	Active
5	Minor
6	Moderate
7	Strong
8	Severe

Table 1. 5 The Geomagnetic storm level Ap and Ae index

Phases	Geomagnetic index	
	Ap	AE
Quiet	≤ 20 nT	≤ 200 nT
Low	≥ 30 nT	≤ 400 nT
Moderate	≥ 50 nT	≤ 800 nT
Strong	≥ 80 nT	≥ 1000 nT
Intense	≥ 100 nT	≥ 2000 nT

1.1.10 Seasonal and Diurnal variations

Diurnal variations are caused by the rotation of Earth, as ionosphere varies between day and night. Photoionization tends to increase during the day and which started to decrease by sunset. Ionosphere varies from season to season because solar zenith angle also influence the TEC that phenomenon implies seasonal variation.

1.8. Different kinds of Earthquake precursors

To reduce the number of loses and prevent disaster EQ prediction has been investigated since long time. Early warning signs before EQ strike are EQ precursors. These warning signs include induced electric or magnetic field,

Ground water level changes, temperature changes[25] , atmospheric and ionospheric anomalies as shown in figure 1.8 [26]. Seismo-ionospheric precursor's study is the growing area of research and achieved considerable importance to measure and identify seismo ionospheric disturbance TEC is the way [27][28][29].GPS TEC assist in studying relationship between EQ and ionospheric precursors[30]. Earliest ionospheric disturbance related to EQ was detected in 1964 after the Alaska EQ (M_w 9.2, March 28, 1964)[31]. The ionospheric anomalies occurred due to EQs, and raised upward to the ionosphere in the form of acoustic gravity waves and vertical electric fields into the ionosphere[32]. Emergence of gases such as Radon from epicenter causes some chemical modifications which prevail up to the ionosphere and cause disturbance[33]. Furthermore, the pre-seismic ionospheric anomaly is also revealed as false signal and it is not considered as EQ precursor[34]. The geomagnetic anomalies in the ionosphere are also due to the extreme solar storm. Therefore, it is required to check geomagnetic indices before and after the EQ. The chief purpose of including geomagnetic indices in thesis is to figure out either ionospheric irregularities are due to geomagnetic storm or EQ. Like many studies, our present study shows that geomagnetic storm is completely quiet four days before EQ but the storm is active beyond five days before and after main shock. So, the anomalies appear due to storm [35][36]. Furthermore, enhancement in TEC prior to EQ supports that perturbations in ionosphere are because of induction of storm over epicenter.

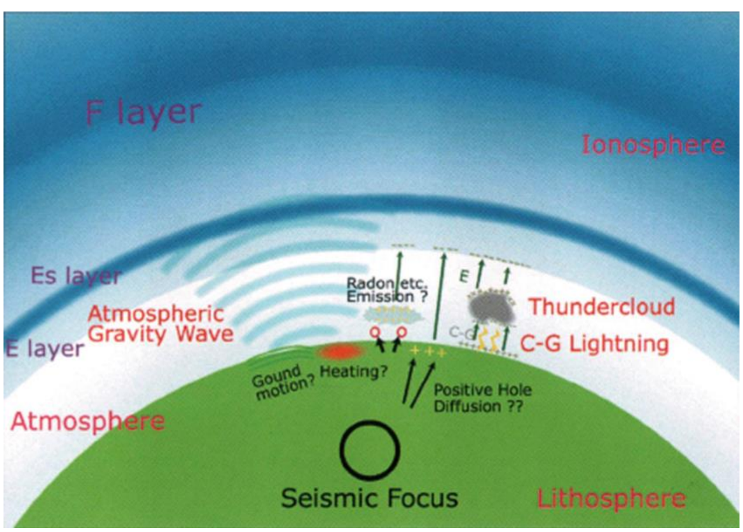

Fig. 1. 9 Seismo-ionospheric precursors

1.9.Total electron content (TEC) as EQ precursor

During EQ preparation time the phenomena involved in upper atmosphere is fairly a new field and gained promising success. Studies are based on GNSS observations, observations include not only navigation but ionospheric studies/upper atmospheric studies. GNSS signals are encountered by ionosphere before receiver. Ionosphere comprised of free electrons that effect the propagation of signals, which are being observed by receiver later[37]. Because of this process GNSS based facilities detect EQ precursors few days before and after EQ because of ionospheric irregularities induced by EQ process. TEC is the parameter of ionosphere which produce effects on signal propagation. TEC is termed as the integral of electron density in a 1 metre square column along the signal transmission path, where 1 TECU = 10^{16} electron/ m^2 [38]. Ionosphere cause signal delay and TEC measurement that are being obtained by GNSS receivers are important parameter to characterize ionosphere. Fourth coming EQ is being detected by change in ionosphere so it’s mandatory to investigate TEC variations because of tectonic distortion prior to an EQ. [39]

2 DATA AND METHODOLOGY

2.1 Case study

2.1.1 Earthquake and database

On January 11, 2021, an EQ occurred at 33 Km SSW of Turt, Mongolia having M_w 6.7, Mongolia EQ occurred at UT = 21: 32: 59 hours at geographical placement upon Latitude (51.21°N) and Longitude (100.47°E) as claimed by United States Geological Survey (USGS). This EQ has a shallow hypo central depth of 10 km, as shown in Fig. 1. There are about 19000 people living within 100 km of epicenter. USGS PAGER estimates that up to 4000 people were exposed to strong shaking https://www.gdacs.org/report.aspx?eventid=1251188&episodeid=1352410&eventtype=EQ. In the present study, the ionospheric anomalies related to with shallow hypo central EQ Mw 6.7 is investigated using the TEC of different IGS stations working in respective seismogenic zone. Various morphological parameters of the EQ include amplitude i.e. the size of the wiggles on an earthquake recording, depth, and location and the data of the EQ is obtained via website of USGS http://eq.usgs.gov/EQs/search/.

2.2 Geomagnetic and storm indices data

Data of geomagnetic storm indices (Kp, Dst, Ap and F10.7) is acquired from International Services of Geomagnetic Indices (ISGI) https://isgi.unistra.fr/datadownload.php.and OMNI web NASA https://omniweb.gsfc.nasa.gov/form/dx1.html.

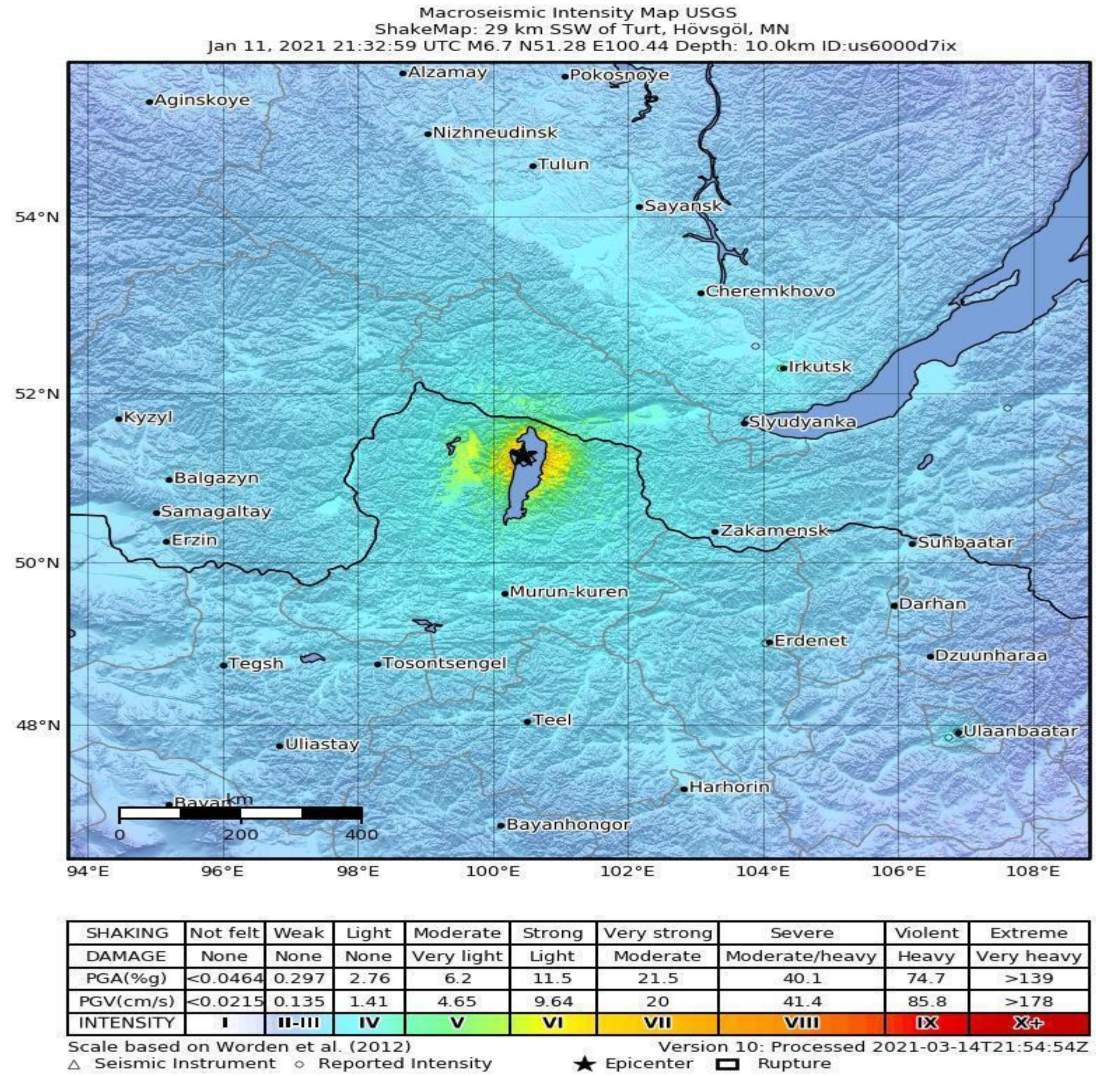

Fig. 2. 1 Location of the EQ epicenter. Black star shows the coordinates of epicenter and colored portion depicts the intensity of EQ

2.3 GNSS TEC data

GNSS contains different constellation system that revolves around the Earth in particular orbits transmits different signals in different bands (carrier waves) provide navigation and position data along with ionospheric information. For accurate positioning ionospheric delays has been removed by ionospheric models.by tracking the phase difference of carrier waves we monitor the temporal changes of TEC along LOS (called slant TEC). TEC is typically defined in TEC units. GNSS TEC assist in studying relationship between EQ and ionospheric precursors. To look into the differences between seismic and storm anomalies during Dec 22-Jan 21, 2021. The IGS stations provide Slant TEC (STEC).

$$\mathrm{STEC} = \frac{f_2^2 f_1^2}{40.3(f_2^2 - f_1^2)}(L_1 - L_2 + \lambda_1(S_1 + b_1) - \lambda_2(S_2 + b_2) + \epsilon) \qquad (2)$$

$$\mathrm{STEC} = \frac{f_2^2 f_1^2}{40.3(f_2^2 - f_1^2)}(P_1 - P_2 - (d_1 - d_2) + \epsilon) \qquad (3)$$

Where STEC denotes Slant Total Electron Content, f_1 (1575.42 MHz) is carrier phase frequency at one end (1x1) square meter tube of GPS signals and f_2 (1227.60MHz) is frequency (f_2) at second end.P_1 and P_2 are the pseudo ranges measured in L1 and L2, respectively. S represents ambiguity of signal and b denotes the instrumental biases, differential code bias is represented by d. Moreover, transformation from Slant TEC into Vertical TEC is done by the succeeding equation [40].

$$VTEC = STEC \times \cos\left(arcsine\left(\frac{RsinZ}{R+H}\right)\right)$$

Where, R symbolizes the radius of Earth and H denotes the single layer height. Likewise, elevation angle of satellite is represented by Z.

To probe into the ionospheric abnormalities and its relationship with the EQ, the TEC data is obtained from permanent three International GNSS Service (IGS) stations functioning in seismogenic zone named as KSTU, ULAB, and IRKT is monitored to compute upcoming EQ relation with ionosphere. Geographical location of TEC stations and their distance from epicenter information is given in Table.2.1

Table 2. 1 Geographical information of 2021 Mw 6.7 Mongolia EQ

S. No.	GPS-TEC Stations	Geographic Coordinates	
		Latitude	**Longitude**
1.	KSTU	55.59	92.47
2.	ULAB	47.51	107.03
3.	IRKT	52.13	104.18

Within EQ breeding zone VTEC is obtained from different stations via method mentioned in [40]. Continuous 20 days before and 10 days after main shock is observed and statistically analyzed which is explained later. The TEC from these stations are obtained from http://www.ionolab.org/index.php?page=iriplasopt&language=en.

2.4 Dobrovolsky region

These aforementioned stations are recording data within the respective EQ breeding zone defined by Dobrovolsky et al.[41] eq. is

$R = 10^{0.43M}$ km.

Radius of breeding zone is represented by R and M symbolizes the magnitude of EQ. Radius of the EQ is calculated by Dobrovolsky equation and it is about 760.3 km as in Fig. 2.2

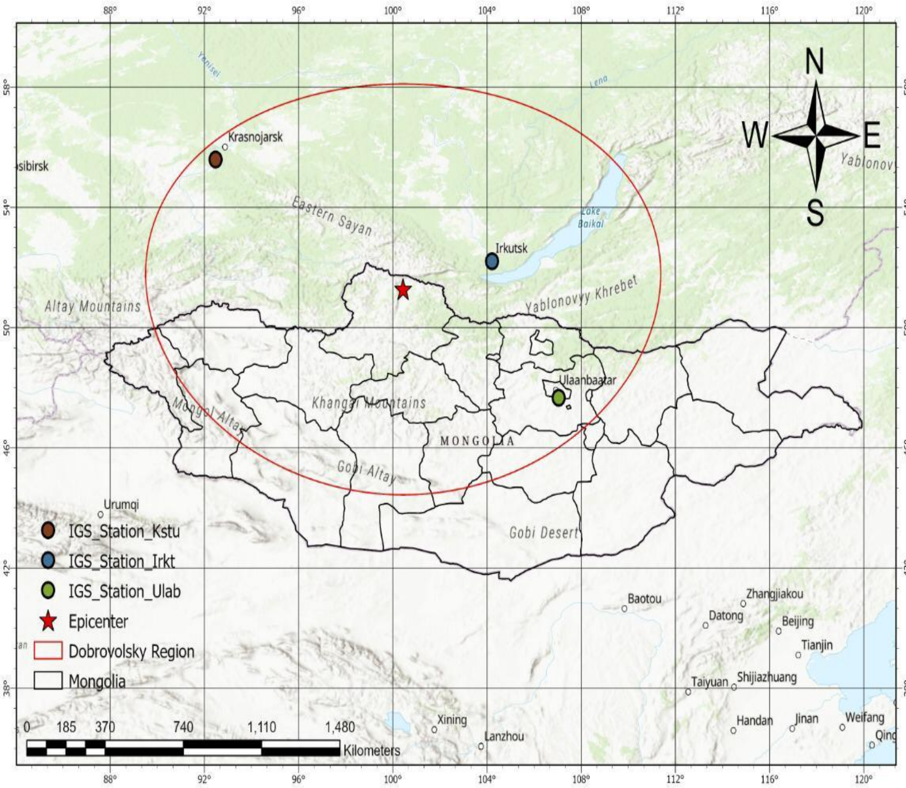

Fig. 2. 2 Location of EQ epicenter and GNSS permanent stations functioning in breeding zone. The red star depicts the site of the epicenter where small colored circles represents IGS stations and red line denotes the EQ breeding region estimated by Dobrovolsky equation.

2.5 Pre seismic ionospheric disturbance

Before larger EQ anomaly that appeared in TEC is termed as Preseismic ionospheric disturbance. To study Preseismic TEC anomalies two different approaches are used that is short term anomalies and long term anomalies. Scientist investigated long-term precursors by observing behavior of GPS TEC 3 to 4 days before EQ and found anomalies before large EQ like in [42] we found that TEC show anomalous behavior few days before EQ.

To figure out TEC long term anomalies we statistically analyze TEC data and thereby, get vertical TEC. we considered anomaly if original TEC is greater or less than upper bound that is further elaborate in next paragraph.

2.6 Statistical estimation of ionospheric TEC disturbance

In present study, I statistically analyzed TEC by obtaining confidence bounds every day 20 days before and ten days after EQ as like previous findings[29][43][19][37]. These are calculated by the mean and standard deviation of twenty pre and ten post days after the EQ by means of the following equation. Then upper and lower bounds are calculated by [27] :

$$\text{Upper Bound} = m + 2\sigma, \qquad (3)$$

$$\text{Lower Bound} = m - 2\sigma, \qquad (4)$$

the deviations in TEC between the observed TEC and upper bound are found pre and post the EQ. Deviation TEC (DTEC) is used to enhance the results of TEC[44][45]. Deviation TEC can be calculated by this formula:

$$TEC_{deviation} = TEC_{original} - \text{upper bound} \qquad (5)$$

Two types of ionospheric anomalies appeared after statistical investigation of data that is positive anomaly and negative anomaly. As per previous results TEC above upper confidence bound is called as positive ionospheric anomaly [35]. Whereas, if GPS-TEC data is below upper confidence bound then it is termed as negative anomaly.

Motivation of current study is to observe if precursory TEC anomalies happened before and after Mw 6.7 EQ of Mongolia. It has been cleared that anomaly appeared is long term anomaly.

3 RESULT AND DISCUSSIONS

3.1 Seismo-Ionospheric Anomalies Associated with Mongolia EQ

In current analysis, seismo ionospheric anomalies are examined from GNSS-TEC associated with the Turt EQ, Mongolia on Jan 11, 2021 having Mw 6.7 with depth 20km and geographical location Latitude (51.21^{o}N) and Longitude (100.47^{o}E) as claimed by United States Geological Survey (USGS) . Changes in the form of anomalies detected twenty days pre and ten days post the EQ. To explore whether ionospheric anomalies appear due to EQ or solar storm geomagnetic storm indices are included to study the effect. Our study is done under quiet storm condition. Table 1.3-1.5 shows geomagnetic indices and their severity. Changes in the form of anomalies detected twenty days pre and ten days post the EQ[46]. Our outcomes depict that geomagnetic storm (Kp>4) induced perturbation five days pre and post the M_w 6.7 EQ, showing no relation with epicenter [47].

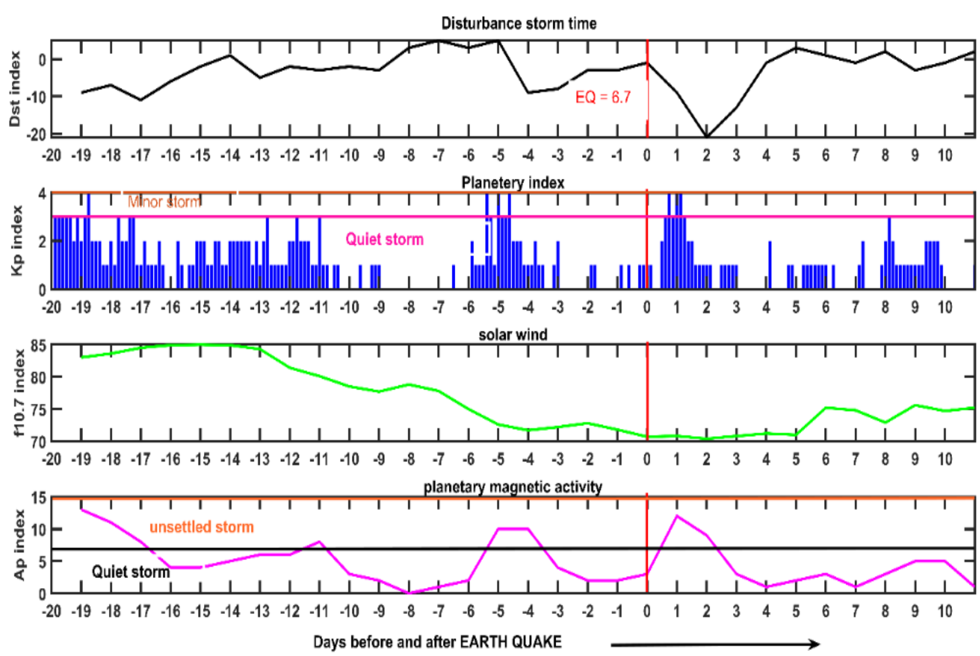

Fig. 3. 1 Geomagnetic storm i.e., Dst, Kp and Ap and F10.7 data have been observed during the EQ period. EQ day is represented by red line.

The storm is active beyond 5 days pre and post the EQ as shown in Fig, 3.1

The TEC retrieved from GNSS stations within the EQ preparation region displays irregularities within five days before the EQ (e.g., -4, -3, -2, -1) [42]. For the detection of anomalies, the TEC for twenty days before and ten days after the main event is computed. To recognize any significant TEC anomalous trends, confidence bounds of daily TEC values are calculated. The limits are calculated by the mean and standard deviation of twenty pre and ten post days after the EQ by means of the equation 3 and 4 as results are depicted in fig.

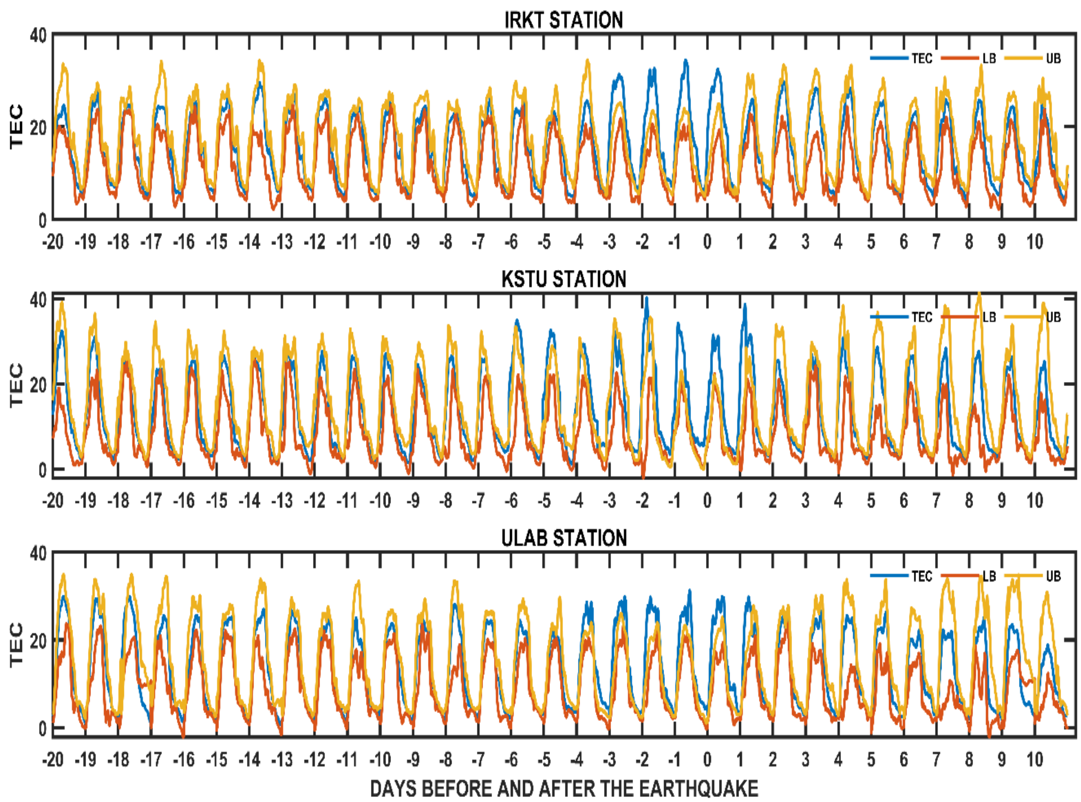

Fig. 3. 2 TEC measurement recorded at three IGS stations (KSTU, ULAB, IRKT)for thirty days twenty days prior and ten days next to EQ.

Similarly, the deviations in TEC between the observed TEC and upper bound are found pre and post the EQ . Enhancement in TEC prior to EQ supports that anomaly may be due to Kp>3 storm over EQ breeding zone. Deviation TEC (DTEC) is calculated to enhance the results of TEC. Deviation TEC can be calculated by the formula shown in eq 5 and results are depicted in fig 7. As per previous results TEC above upper confidence bound is called as positive ionospheric anomaly. Fig.7 depicts DTEC anomalies that are because of geomagnetic storm and no association of activating ionospheric disturbance has been observed[48][47].

The cross wavelet power spectrum of occurrence rate of TEC anomalies 20 days before and 10 days after main shock. It shows that increase in TEC disturbances appeared at 4th day before main event. After 4th day, the disturbances were prolonged and enlarged, and attained its maximum value at EQ day. Then, the TEC disturbances disappeared after 2 days of main shock. Our findings indicate that abnormal disturbances of ionospheric TEC lie between -4, -3, -2, -1 and day of EQ. Further, storm is active 5 days before main shock and right after it so anomalies in ionosphere is induced by storm as shown in figure 8[49].

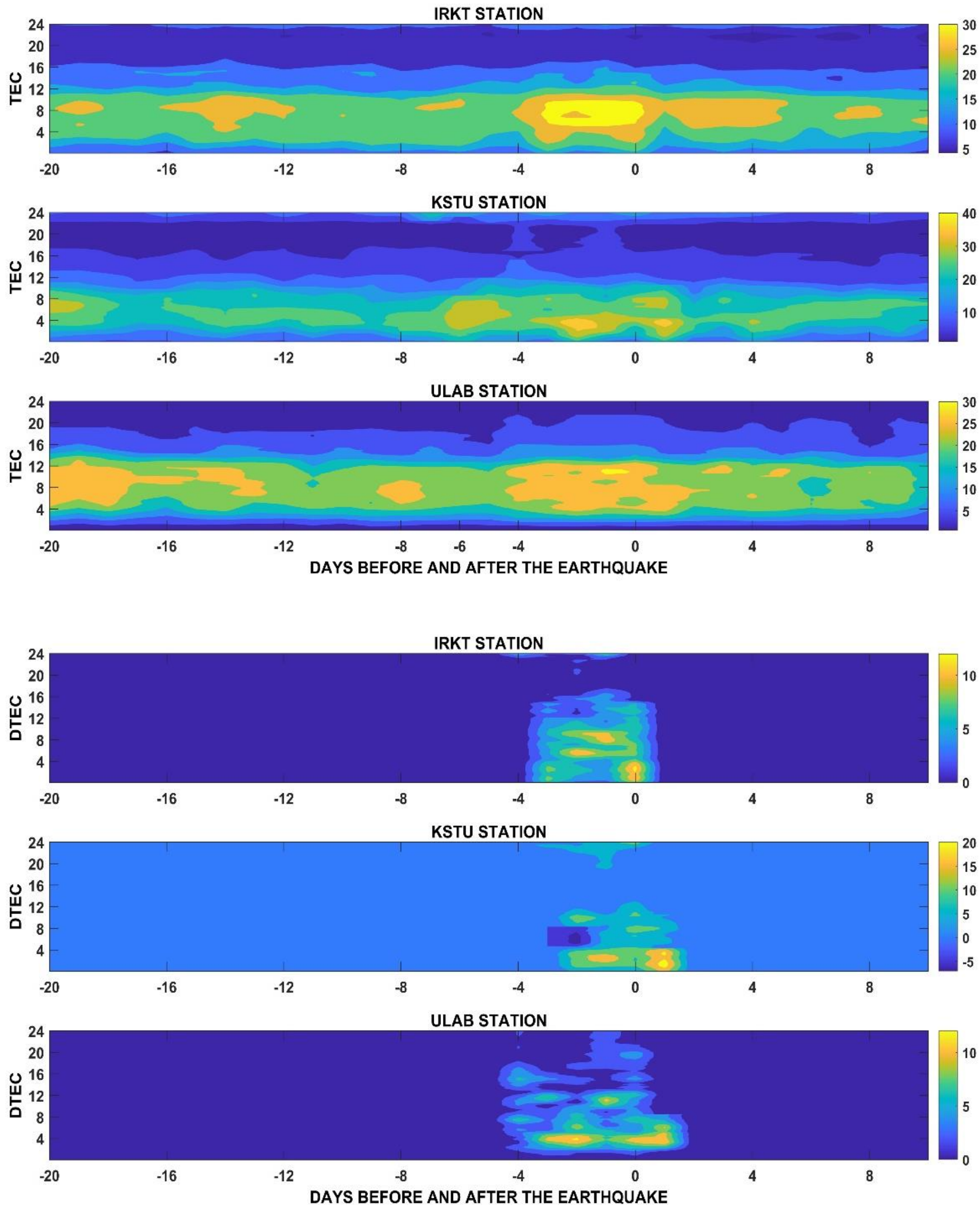

Fig. 3. 3 Spatial TEC (upper) and DTEC (lower) recorded at three IGS stations (KSTU, ULAB, IRKT) shows variations due to geomagnetic storm twenty days before and ten days after EQ.

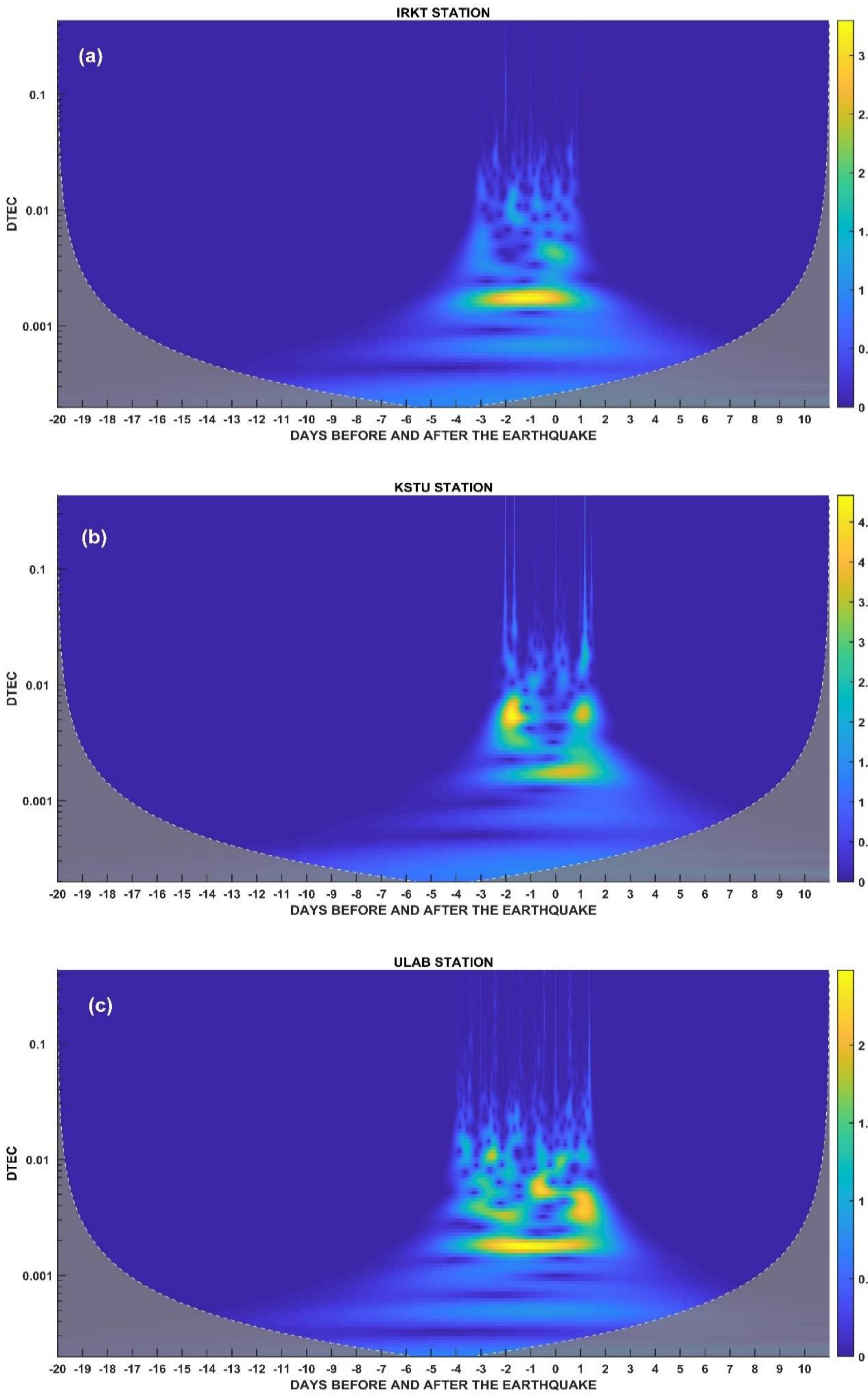
IRKT STATION
(a)
KSTU STATION
(b)
ULAB STATION
(c)
DTEC
0.1
0.01
0.001
DAYS BEFORE AND AFTER THE EARTHQUAKE
-20 -19 -18 -17 -16 -15 -14 -13 -12 -11 -10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10

Fig. 3. 4 Wavelet spectrum of DTEC retrieved from IGS station (a) IRKT (b) KSTU and (c) ULAB stations depicts TEC disturbances before main event of EQ.

3.2 CONCLUSION

In the presented research, seismo ionospheric perturbations of shallow hypo central EQ having Mw 6.7 monitored by IGS stations in Mongolia functioning nearby epicenter from GPS TEC, has been discussed. The geomagnetic storm is active beyond 5 days before EQ of Mw 6.7 i.e., (Kp>4) it show no association with epicenter. Investigation on TEC improve our results which depicts that enhancement and depletion in TEC prior to EQ supports the induction of storm anomalies over the seismogenic zone.

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